



The Department of Mathematics and Computer Science

About

Statistics

Number Theory

Java

Data Structures

Cornerstones

Calculus

Triangle of Power Notation and Logarithms

When it comes to the relationship $a^b = c$, we have examined how c can be thought of as a combination of a and b . We have also considered the implications of thinking of a as a combination of b and c . There is one more possible combination we could contemplate -- what happens if we view b as a combination of a and c ?

John Napier considered something similar in 1614, calling such combinations "logarithms" (a term he invented by jamming together the Greek words for "proportion" (i.e., *logos*) and "number" (i.e., *arithmos*).

In modern mathematics, we abbreviate the value resulting from the combination of a and c that will yield b above with $\log_a c$. (Note, this presumes $a \neq 1$ is positive, as otherwise there will be problems.) That is to say:

$$a^b = c \iff \log_a c = b$$

Recall we have previously referred to the a in the equation on the left as the "base" of the power a^b . In a similar way, we call the a in the equation on the right the **base** of the logarithm $\log_a c$, which we then read as "the log base a of c ".

*Often in the sciences and elsewhere, folks must deal with very large or very small values and employ scientific notation, which uses powers with a base of 10, to express these. As such, it is not uncommon for texts to adopt the convention of writing $\log x$ when they really mean $\log_{10} x$. However, one must be careful! In computer science, where data is represented in "bits" which take on two states, powers with a base of 2 are more common. In texts in this discipline, one may find $\log x$ meant to be $\log_2 x$. In still other disciplines, yet another value (Euler's number, e) will be more "natural" in calculations, and $\log x$ will be used with this base to be implied instead. The upshot of this is that when the base is not written, it is assumed to be the base most common to the discipline in question. As such, when we write $\log x$, we say we have taken the **common log** of x .*

We can -- and will -- explore the properties associated with these logarithms. However, before doing so, let us pause for a moment and consider the beautiful symmetry connecting these three ideas: powers, roots, and logarithms

... a symmetry some have enshrined in a new notation known as the "**Triangle of Power**".

As far as notations go, the "triangle of power" is admittedly so young that one could conceive of it as still gestating in the womb! It made its first appearance on Stack Exchange in July of 2012, but most who know of it now do so due to Grant Sanderson endorsing it four years later on his YouTube channel [3Blue1Brown](#) which is dedicated to visualizing mathematical ideas through beautiful animations -- a channel that boasts 6.44 million subscribers as of 2024, by the way!

As new and as-yet-unaccepted by the general mathematics community as it still is -- once seen -- many will agree it at least meets the criteria for good notation established in the following quotation by famed British logician Bertrand Russell:

"A good notation has a subtlety and suggestiveness which at times make it almost seem like a live teacher. -- B. Russell"

Here's how triangle of power notation works: First, we express the relationship $a^b = c$ by writing: $\triangle_{a,c}^b$

This triangle then serves as a basis for denoting each of the three values at its vertices as a combination of the other two. We do this by duplicating the triangle above, but leaving the value the combination equals missing.

Consider how much cleaner and more symmetric the results appear (see the statements below on the left) as compared to the same facts written with more traditional notation (on the right):

$$\begin{array}{ccc} \triangle_{a,c}^b = c & \longleftrightarrow & a^b = c \\ \triangle_{a,c}^b = b & \longleftrightarrow & \log_a c = b \\ \triangle_{a,c}^b = a & \longleftrightarrow & \sqrt[b]{c} = a \end{array}$$

So that we can manipulate such expressions without first "translating back" to our traditional notations, note how some of our familiar rules for exponents look in "triangle of power" notation:

$$x^m \cdot x^n = x^{m+n} \quad \longleftrightarrow \quad x \triangle_x^m \cdot x \triangle_x^n = x \triangle_x^{m+n}$$

$$\frac{x^m}{x^n} = x^{m-n} \quad \longleftrightarrow \quad \frac{x \triangle_x^m}{x \triangle_x^n} = x \triangle_x^{m-n}$$

$$(xy)^n = x^n y^n \quad \longleftrightarrow \quad xy \triangle_{xy}^n = x \triangle_x^n \cdot y \triangle_y^n$$

$$\left(\frac{x}{y}\right)^n = \frac{x^n}{y^n} \quad \longleftrightarrow \quad \frac{x \triangle_x^n}{y \triangle_y^n} = \frac{x \triangle_x^n}{y \triangle_y^n}$$

$$(x^m)^n = x^{mn} \quad \longleftrightarrow \quad x \triangle_x^m \triangle_{x \triangle_x^m}^n = x \triangle_x^{mn}$$

Admittedly, the new notation doesn't really seem to look any better for exponents. Indeed, it actually looks more cumbersome here, in that we are using more "ink" to express the same ideas. That said, notice how the following rules for radicals (for appropriate values of x , y , m , and n) compare:

$$\sqrt[n]{x} \cdot \sqrt[n]{y} = \sqrt[n]{xy} \quad \longleftrightarrow \quad \triangle_{xy}^n = \triangle_x^n \cdot \triangle_y^n$$

$$\sqrt[n]{\frac{x}{y}} = \frac{\sqrt[n]{x}}{\sqrt[n]{y}} \quad \longleftrightarrow \quad \triangle_{\frac{x}{y}}^n = \frac{\triangle_x^n}{\triangle_y^n}$$

$$\sqrt[n]{\sqrt[m]{x}} = \sqrt[nm]{x} \quad \longleftrightarrow \quad \triangle_{\triangle_x^m}^n = \triangle_x^{mn}$$

Here we start to see the niceties of this notation -- look how symmetric the 3 results above look in comparison to the last 3 results we gave for exponents!

Inverse Relationships

The symmetry doesn't stop there. Recall how exponents, roots, and logarithms can sometimes undo one another: When ***a*** and ***b*** are positive, we have

$$\sqrt[a]{b^a} = b \quad \leftrightarrow \quad \triangle_b^a \triangle_a^a = b$$

$$(\sqrt[a]{b})^a = b \quad \leftrightarrow \quad \triangle_b^a \triangle_a^a = b$$

$$a^{\log_a b} = b \quad \leftrightarrow \quad \triangle_a^a \triangle_a^b = b$$

(In case the last relation wasn't obvious, recall $a^x = b \iff \log_a b = x$, and then substitute the ***x*** in the first with the $\log_a b$ of the second.)

Looking at the structure of the triangle notation seen in the last three examples, notice how each large triangle has an ***a*** at one vertex with the smaller triangle having an ***a*** and ***b*** in the two positions closest to the big triangle. Then, in each case we can "cancel" both values of ***a***.

Kinda' makes one wonder what happens when we see this same structure in the other ways it could occur, right? Wouldn't it just "feel right" if the following were all equal to ***b*** too?

$$\triangle_a^a \triangle_a^b = b \quad \triangle_b^b \triangle_a^a = b \quad \triangle_b^a \triangle_a^a = b$$

But this would mean all the following are true, respectively...

$$\log_a a^b = b \quad \log_{\sqrt[a]{a}} a = b \quad (\log_b a) \sqrt[a]{a} = b$$

They are!

The first of the three above (as well as the last of the three before these) is traditionally taught to students as one of the "inverse relationships" for logs and is normally justified like this: We know $a^b = c$ is equivalent to $\log_a c = b$. In this way, we can think of $\log_a c$ as "*the exponent needed on a to produce a power equal to c* ". As such, we can interpret $\log_a a^b$ as "*the exponent needed on a to produce a^b* ", which is of course b .

Likewise, the earlier seen $a^{\log_a b}$ can be interpreted as "*the value a raised to precisely the power needed to raise a to get b* ". Consequently, we not surprisingly get b .

The last two of the three equations above are inverse relationships not as often seen -- but every bit as valid! [Note, the reader will benefit from taking a moment to convince themselves these two equations are also true.](#)

Properties of Logarithms (Logs of Products / Sums of Logs)

The "triangle of power" notation has more to tell us about logarithms, revealing even more symmetries...

Suppose ${}_x\Delta_a^m$ and ${}_x\Delta_b^n$ for positive x , a , and b . (That is to say, suppose $x^m = a$ and $x^n = b$.)

As a consequence,

$$ab = {}_x\Delta_a^m \cdot {}_x\Delta_b^n = {}_x\Delta^{m+n}$$

But then,

$$\begin{aligned} {}_x\Delta_{ab} &= {}_x\Delta^{m+n} && \text{after substituting } {}_x\Delta^{m+n} \text{ for } ab \\ &= m + n && \text{after "canceling" the } x, \text{ as discussed earlier} \\ &= {}_x\Delta_a^m + {}_x\Delta_b^n && \text{after rewriting } m \text{ and } n \text{ in triangle of power notation} \end{aligned}$$

As such, we have

$${}_x\Delta_{ab} = {}_x\Delta_a + {}_x\Delta_b$$

In traditional notation, we just proved the perhaps more familiar result that for positive x , a and b :

$$\log_x ab = \log_x a + \log_x b \quad (\text{i.e., the log of a product is a sum of logs})$$

Again, notice the symmetry between this and the exponent rule for products of powers of the same base::

$$\begin{aligned} {}_x\Delta_{ab} = {}_x\Delta_a + {}_x\Delta_b &\longleftrightarrow \log_x ab = \log_x a + \log_x b \\ {}_x\Delta^{a+b} = {}_x\Delta^a \cdot {}_x\Delta^b &\longleftrightarrow x^{a+b} = x^a \cdot x^b \end{aligned}$$

Properties of Logarithms (Logs of Quotients / Differences of Logs)

A similar argument using "triangle of power" notation establishes the following logarithmic

relationship (i.e., the log of a quotient is a difference of logs).

Again suppose ${}_x\Delta_a^m$ and ${}_x\Delta_b^n$ for positive x , a , and b . Then,

$$\frac{a}{b} = \frac{{}_x\Delta_a^m}{{}_x\Delta_b^n} = {}_x\Delta^{m-n}$$

But then

$$\begin{aligned} {}_x\Delta_{\frac{a}{b}} &= {}_x\Delta_{{}_x\Delta^{m-n}} \\ &= m - n \\ &= {}_x\Delta_a - {}_x\Delta_b \end{aligned}$$

As such, we have

$${}_x\Delta_{\frac{a}{b}} = {}_x\Delta_a - {}_x\Delta_b$$

In traditional notation, we just proved the following important result for positive x , a , and b :

$$\log_x \frac{a}{b} = \log_x a - \log_x b \quad (\text{i.e., the log of a quotient is a difference of logs})$$

Note the symmetry this logarithmic rule has with the rule of exponents that directs us to subtract exponents when dividing powers of the same base, as shown:

$${}_x\Delta_{a \div b} = {}_x\Delta_a - {}_x\Delta_b \quad \longleftrightarrow \quad \log_x \frac{a}{b} = \log_x a - \log_x b$$

$${}_x\Delta^{a-b} = {}_x\Delta^a \div {}_x\Delta^b \quad \longleftrightarrow \quad x^{a-b} = \frac{x^a}{x^b}$$

Properties of Logarithms (Logs of Powers and Multiples of Logs)

Of course, once we have ${}_x\Delta_{ab} = {}_x\Delta_a + {}_x\Delta_b$, the suggestion that ${}_x\Delta_{a^n} = n({}_x\Delta_a)$ is immediate given the obvious pattern appearing below:

$$\begin{aligned} {}_x\Delta_{a^2} &= {}_x\Delta_a + {}_x\Delta_a = {}_x\Delta_a + 1({}_x\Delta_a) = 2({}_x\Delta_a) \\ {}_x\Delta_{a^3} &= {}_x\Delta_a + {}_x\Delta_{a^2} = {}_x\Delta_a + 2({}_x\Delta_a) = 3({}_x\Delta_a) \\ {}_x\Delta_{a^4} &= {}_x\Delta_a + {}_x\Delta_{a^3} = {}_x\Delta_a + 3({}_x\Delta_a) = 4({}_x\Delta_a) \\ &\vdots \end{aligned}$$

While the pattern above leads naturally to a proof (by induction[†]) of the suggested result, such an argument would only establish it for positive integers n . However, we can do better.

[†] If you are curious what this "induction" thing is all about, check out the following notes on the subject: [The Principle of Mathematical Induction](#) and [Variations on Induction](#). We don't need to know about this important means of mathematical argument just yet -- but we will eventually.

The following more direct (and shorter) argument will establish the same, but for all real values n :

Suppose $y = {}_x\Delta_a$. Then $x^y = a$. Raising both sides to the n^{th} power, we have $(x^y)^n = a^n$. But then, $x^{ny} = a^n$, so ${}_x\Delta_{a^n} = ny$. Finally, swapping out y with ${}_x\Delta_a$ gives us the desired result:

$${}_x\Delta_{a^n} = n \left({}_x\Delta_a \right)$$

Of course, we must acknowledge the calculations immediately above actually "mix" traditional and triangle-of-power notations. For the triangle-of-power purists out there, note that we could also write our result in this form:

$${}_x\Delta_a \overset{n}{\Delta} = n \left({}_x\Delta_a \right)$$

Still, it is perhaps easier to see the parallels between this result and the traditionally-written "log-of-a-power" rule for logarithms if we use the aforementioned "mixed form":

$${}_x\Delta_{a^n} = n \left({}_x\Delta_a \right) \iff \log_x a^n = n \log_x a \quad (\text{the log of a power is multiple of a log})$$

There is a parallel result when the base is a power. Suppose $\log_{b^p} x = y$. Then,

$$b^p \overset{y}{\Delta} x \text{ and hence, } b \overset{py}{\Delta} x \text{ which implies } \log_b x = py$$

Dividing both sides by p and noting the two expressions for y must of course be equal, we conclude

$$\log_{b^p} x = \frac{1}{p} \log_b x \quad (\text{exponents associated with the base can be "pulled out" as reciprocal multipliers})$$

The "Change of Bases" Theorem for Logarithms

Another important theorem for logarithms also follows quickly from the exponent rule whereby $(a^x)^y = a^{xy}$. To see this, consider the following:

Suppose ${}_a\Delta_b^x$ and ${}_b\Delta_c^y$ for positive a , b , and c (that is, $a^x = b$ and $b^y = c$)

But then,

$$\begin{aligned} c &= {}_b\Delta^y \\ &= {}_a\Delta^x \overset{y}{\Delta} \quad \text{after substituting } {}_a\Delta^x \text{ for } b \\ &= {}_a\Delta^{xy} \quad \text{after applying the "triangle of power" variant of } (a^x)^y = a^{xy} \end{aligned}$$

The conclusion can be written in multiple ways:

$$c = {}_a \Delta^{xy} \longleftrightarrow {}_a \Delta_c^{xy} \longleftrightarrow a^{xy} = c$$

Focusing on the middle one, it must be that

$$\begin{aligned} {}_a \Delta_c &= xy \\ &= {}_a \Delta_b \cdot {}_b \Delta_c \end{aligned}$$

Dividing both sides by ${}_a \Delta_b$ yields

$$\frac{{}_a \Delta_c}{{}_a \Delta_b} = {}_b \Delta_c$$

or again, in the more traditional notation -- the "Change of Bases" rule for logarithms:

$$\frac{\log_a c}{\log_a b} = \log_b c$$

The theorem bears this name as it can be used to change the base of any and all logarithms that might appear in an expression to any desired legal value.

This is particularly important when one needs to evaluate a logarithm on a calculator. Depending on the sophistication of one's calculator, you might only have a single "log" button that finds **log x** with an assumed "common base" of **10**.

Armed with only this, what if you need to find the value of **log₂ 7 + log₃ 5**? Do you need a fancier calculator? Certainly not!

With the change of bases theorem as stated above, and using **a = 10**, we can rewrite the value we seek:

$$\log_2 7 + \log_3 5 = \frac{\log_{10} 7}{\log_{10} 2} + \frac{\log_{10} 5}{\log_{10} 3}$$

which could then be typed into the calculator as

$$\log(7)/\log(2) + \log(5)/\log(3)$$

The "Simple-to-Complex" Convention for Writing Products and Variations on Notations for Logarithmic Expressions

Consider the following two products:

$$2xy(x+6)\sqrt{x^5-3x+9} \quad \text{vs.} \quad \sqrt{x^5-3x+9}xy(x+6)2$$

For any given pair of values for **x** and **y**, these two expressions absolutely calculate the same value. However, the "simple-to-more-complicated" ordering of the factors in the left expression has some significant advantages over the ordering used on the right.

First, note that if one writes the expression on the right "by hand", one can easily accidentally overextend the vinculum on the radical (i.e., the bar at the top) to a point where it appears the **x** (or maybe even both **x** and **y** depending on how sloppy one's handwriting might be) is *inside* the radical! This of course, would be a very different expression!

Similarly, if we wrote the **2** on the end of the second expression a bit higher than intended, it might be erroneously interpreted as squaring something instead of doubling something.

The ordering in the expression on the left avoids these two pitfalls.

Therefore, having a default of writing things in a "simple-to-complex" form as we move from left to right in an expression is a good idea (*as long as doing so is legal -- remember a lack of commutativity would cause problems here*) -- and is a general convention followed by mathematicians. [The reader is strongly encouraged to adopt this convention too!](#)

When one can be assured this convention is being employed, there are some consequences for how expressions involving logarithms are interpreted.

Using the traditional notation, logarithms are sometimes written with parentheses and sometimes without. For example, the following are unambiguously equivalent

$$\log_2 8 = \log_2(8)$$

When the value of the "power" involved (e.g., the **8** above) is some explicit number or single variable, the form on the left is often preferred (as it uses less ink).

However, consider the expression $\log_2 4 + 4$. Should this be thought of as $\log_2(4 + 4)$ or $(\log_2 4) + 4$? As the first equals **3** and the second equals **6**, it clearly makes a difference! In such cases we really need to parentheses to clear up any ambiguity.

In the spirit of ink-saving, some will also write $\log_7 ab$ when they mean $\log_7(ab)$. This is not unreasonable, for a couple of reasons: First, the **b** looks to be drawn quite close to the **a**, whereas the space between the the $\log_7$ and the **a** appears slightly larger -- suggesting by their proximity a "grouping" together of these two factors, which then results in prioritizing their combination before others. Second, when we *assume* the expression has been written so that smaller, simpler factors before larger, more complicated ones -- we conclude the **b** must be grouped together with the **a** to form the "power" part of the logarithm, as otherwise the **b** is far too simple in comparison with the factor $(\log a)$ to be on its right!

Great care needs to be exercised here though!

First, one must be certain the assumption that things have been written in a "simpler-to-more-complicated" way from left to right is a good one! If we saw the perfectly valid expression below, we might start to doubt that fact:

$$\log_2 ab + (c + d)\sqrt{x^2 - y}7x(a - d)$$

Second, where does one draw the line? In the five expressions that follow, which expressions are "simple enough" or "spaced tightly together enough" that we should interpret them as being part of the respective logarithm, and which are not -- meaning we should instead treat them as being "outside" the logarithm and a factor of the entire expression?

$$\log_5 ab, \quad \log 4xy^3z^2w^5, \quad \ln x(x + 1), \quad \ln y(y + 1) \quad \log_3 a \left(1 + \frac{1}{n}\right)$$

Again, to be clear about what you intend, include appropriate parentheses. Notice how any confusion about what is meant by the above dissipates upon the addition of parentheses:

$$\log_5(ab), \quad \log(4xy^3z^2w^5), \quad \ln(x(x + 1)), \quad (\ln y)(y + 1), \quad (\log_3 a) \left(1 + \frac{1}{n}\right)$$

Of course, if we go ahead anyways and play this game of saving a few drops of ink by leaving off parentheses (*which we will still sometimes do, so that you will understand such abbreviated notation should you encounter it elsewhere*) -- there are other ramifications as

well.

For example, suppose we want to write the square of the expression $\log_3 x$.

Writing $\log_3 x^2$ is definitely *not* what we want to write in this case, as folks might misinterpret this to be $\log_3(x^2)$. We could certainly write $(\log_3 x)^2$ without creating any confusion. That said -- as a "less-ink" but still unambiguous alternative, many will instead write $\log_3^2 x$. This last form has the advantage of having no parentheses but being clearly distinguishable from $\log(x^2)$.

Of course, we can extend this useful variation in notation beyond just squares. In general, we equate the following for any reasonable value of n :

$$\log_b^n x = (\log_b x)^n$$

The "Natural" Base

There is a particular base for a logarithm that will prove very "natural" to use later -- namely the value of $e \approx 2.7182818284\dots$.

I know what you are thinking -- "What in the world is "natural" about this base?!" The answer to that question is extremely interesting and lies at the nexus of trigonometry and non-real solutions to quadratic equations! With a promise to reveal the details behind this amazing number after we've gotten a little bit more under our belt (e 's origin story involves hyperbolas of all things!) -- let us for now just be aware that this base is so frequently used, that it has its own "ink-saving" notation.

When we intend to write a logarithm where e is the base, such as $\log_e x$, we normally instead write $\ln x$ (short for the Latin *logarithmus naturalis*), reading this (in English) as the "natural log of x ".

All of the properties we have seen for more general logarithms of course hold for natural logs as well:

- $\ln e^x = x$
- $\ln x + \ln y = \ln xy$
- $\ln x - \ln y = \ln \frac{x}{y}$
- $\ln x^n = n \ln x$
- $\log_b a = \frac{\ln a}{\ln b}$ (Change of Bases)

and so on..

Irrational Values Decimal Expansions Don't Terminate or Repeat

Recall that irrational values are simply values that are *not* rational. That is to say, they are values that are *not* writable as a ratio/fraction of two integers.

You might have been told at some point that irrational values when written as decimal values don't terminate or repeat. By "don't terminate", we mean we can't write them with a finite number of digits. By "don't repeat", we mean there is no sequence of digits in the decimal expansion that (at some point) simply occurs over and over again, without end.

Note that the value $1.2373496496496\dots$, starting with the fifth decimal digit appears to

repeat the sequence of digits "**496**" over and over. The suggestion is that because of this (presumed infinite) repetition, the resulting value must be rational.

Putting off for the moment how we might know that, we must acknowledge that without complete knowledge of *all* of the digits past some given point in a number, we don't really know if there really is the *infinite* repetition we say we need, or if there is just some initial repetition of some digit sequence some number of times that eventually breaks down. The " $\dots$ " present above only means there are more digits past the last digit shown. The number **1.23734496496496** $\dots$ could easily be **1.2373496496496859** $\dots$ when presented with more precision!

There is a special notation however, that we can use to let the reader know that we really do mean the value where a given sequence of digits keeps repeating from a given point in its decimal expansion onwards. In this notation, we simply draw a line over the repeating sequence. So for example,

$$1.\overline{2373496} = 1.2373496496496\dots \text{ (where the pattern 496 on the end continues } ad \text{ infinitum)}$$

Now let us return to the suggestion that numbers that don't repeat in this way -- numbers that *can't* be written in this "overline" notation -- must be irrational. How do we know that to be true?

Surprisingly, the answer is a simple one: For any number that *can* be written in this overline notation, there is an easy method/algorithm for rewriting these as fractions of integers (i.e., rational values).

Consider the following demonstration of this algorithm on the number **1.2373496** $\overline{}$ mentioned above:

1. First, let $x = 1.\overline{2373496}$
2. Let d be the number of digits in the repeating sequence. Here $d = 3$. Then, multiply both sides of the equation above by **10** exactly d times. This results in $1000x = 1237.\overline{3496496}$, in this case.
3. Subtract the first equation from the second. That is to say, note that:

$$\begin{array}{rcl} 1000x & = & 1237.\overline{3496496} \\ x & = & 1.\overline{2373496} \\ \hline 999x & = & 1236.1123 \end{array}$$

4. Multiply both sides by **10** enough times to eliminate any digits to the right of the decimal point, if needed. Here, we need to multiply by **10** four times to find

$$9990000x = 12361123$$

5. Finally, divide both sides by the number to the left of x to find

$$x = \frac{12361123}{9990000}$$

As we have shown $x = 1.\overline{2373496}$ is representable as a fraction of integers, x must be rational. A similar argument can be constructed for *any* decimal sequence that has an infinitely repeating digit sequence in this way. Thus, irrational values must necessarily have digits that *don't repeat*.

The argument that irrational values' decimal expansions don't terminate (i.e., they have a finite number of digits) is even easier to show. Again, consider the following example. Suppose $y = 1.287$, a terminating decimal expansion with only **3** digits to the right of the

decimal. We first multiply both sides by **10** enough times to clear the decimal values. Here, that gives us **1000y = 1287**. Then, we divide both sides by the value to the left of the **y** to reveal

$$y = \frac{1287}{1000}$$

Again, seeing a fraction for this (and any other terminating decimal expansion), we know such values are rational. Thus, for a value to irrational requires its decimal expansion not terminate.

*As an interesting addendum to the above, the method given for turning a repeating decimal into a fraction can also be used to establish that $0.999\overline{9} = 9/9 = 1$. This means that $0.999\overline{9}$ and **1** are not just really, really close to one another -- they are the exact same value!*

Some Logarithm Values are Irrational

Playing around with the `log` button on your calculator a bit might lead one to wonder if a lot of logarithm values are irrational. They often have "long and messy" looking decimal values that don't seem to terminate or contain any sequence of digits simply repeated over and over.

As has been said previously however, if you only have access to the first 14 digits or so of the values in question (calculators can only show you so many digits, after all) -- there is no way to tell if the value terminates or starts repeating some time after that!

However, there is another way we could argue that some logarithm values are irrational...

Most will believe from their experience[‡] that prime factorizations of positive integers are unique (i.e., that is to say, there is only one way to break down any positive integer into a product of primes, up to the order of the factors in question).

For example, **120 = 2 · 2 · 2 · 3 · 5**. We could reorder the factors given the commutative property of real numbers under multiplication -- but that's not telling us anything new. As such, re-ordering the factors does not result in a new prime factorization of a number. The important part of this is that there will *never* be a different prime factorization of **120**. When **120** is broken down into a product of primes, there will *always* be exactly **3** twos, **1** three, and **1** five.

A similar statement (with different numbers of course) can be made for *every* positive integer.

‡ : As with most claims in mathematics, we don't have to just rely on our experiences and previous observations to tell us this -- we can prove that prime factorizations are unique. We won't do that here as it takes us too far afield from our content of interest -- but interested students might look up the celebrated "Fundamental Theorem of Arithmetic" in a textbook on "Number Theory".

Now, consider the value $x = \log_2 3$. Suppose the value *was* rational. Then, we could write x as a ratio/fraction of two integers, right? In other words, we could find integers p and q so that

$$\log_2 3 = \frac{p}{q}$$

If this is true however, consider what happens when we multiply both sides by q and then apply the logarithm rule that involves "logs of powers and multiples of logs":

$$q \log_2 3 = p \longrightarrow \log_2 3^q = p$$

Now convert the equation on the right to exponential form:

$$2^p = 3^q$$

Think about what this says, however. Suppose y is the common (positive integer) value shared by 2^p and 3^q above.

Given that $y = 2^p$ and noting 2 is prime, the prime factorization of y must be $\underbrace{2 \cdot 2 \cdots 2}_{p \text{ times}}$

But $y = 3^q$ and 3 is prime as well -- so we know the prime factorization of y must (*also?*) be $\underbrace{3 \cdot 3 \cdots 3}_{q \text{ times}}$

Given what we know about prime factorizations (i.e., one never has more than one for any given positive integer), this is clearly impossible! We have run into a contradiction!

Thus, our original assumption that $\log_2 3$ was rational must have been incorrect.

Clearly, $\log_2 3$ must be irrational!

Nice, right?

Approximating Logs

We can numerically approximate logarithm values (without using the `log` button on our calculator) by taking advantage of the various properties of logarithms mentioned earlier in this section.

As an example, suppose we wish to approximate $\log 7532$ (presuming the "common log" in this instance involves a base of 10):

It will be useful to have approximations for a few other common log values to this end. Although some sets of values are more useful than others.

Since we just discussed prime factorizations and $7532 = 2^2 \cdot 7 \cdot 269$, suppose we knew $\log 2 \doteq 0.3010$, $\log 7 \doteq 0.8451$, and $\log 269 \doteq 2.4298$.

Then

$$\begin{aligned} \log 7532 &= \log(2^2 \cdot 7 \cdot 269) \\ &= \log 2^2 + \log 7 + \log 269 \\ &= 2 \log 2 + \log 7 + \log 269 \\ &= 2(0.3010) + (0.8451) + (2.4298) \\ &= 3.8769 \end{aligned}$$

That said, one might wonder how we find the initial approximations for **log 2**, **log 7**, and **log 269**, since the numbers involved are prime and don't break down any further?

To address these, let us consider the following related (but different) technique for approximating a logarithm:

Suppose we again want to find **log 7532**. We should note right away that the value we seek will be between **3** and **4** as $10^3 \leq 7532 \leq 10^4$. To see this, note that every integer with d digits will be between 10^{d-1} and 10^d .

Thus, **log 7532** = **log**($10^3 \cdot 7.532$) = **3** + **log 7.532**.

It remains to find **log 7.532**.

Recall that we have a way (i.e., the Babylonian method) to approximate square roots. Noting that a square root of a square root is a fourth root, and a square root of a fourth root is an eighth root, etc -- we have a way to approximate $\sqrt{10}$, $\sqrt[4]{10}$, $\sqrt[8]{10}$, etc. Written with rational exponents instead, this means we can approximate $10^{1/2}$, $10^{1/4}$, $10^{1/8}$, etc.

Avoiding the tedium of these calculations, let us simply list the values that result:

$10^{1/2}$	$\doteq$	3.16227766
$10^{1/4}$	$\doteq$	1.77827941
$10^{1/8}$	$\doteq$	1.333521432
$10^{1/16}$	$\doteq$	1.154781985
$10^{1/32}$	$\doteq$	1.074607828
$10^{1/64}$	$\doteq$	1.036632928
$10^{1/128}$	$\doteq$	1.018151722
$10^{1/256}$	$\doteq$	1.009035045
$10^{1/512}$	$\doteq$	1.004507364
$10^{1/1024}$	$\doteq$	1.002251148
$\vdots$		

Now, see what happens when you divide the **7.532** we have left by the $10^{1/2} \doteq \mathbf{3.16227766}$ found above. If the result is ≥ 1 , write down $10^{1/2}$ and update "what is left" to be this new quotient. If not, try the next power of **10** in our list (i.e., $10^{1/4} \doteq \mathbf{1.77827941}$) and do the same. Here, the **7.532** we had left gets updated to become **2.81827534**.

In the rare case that we get **0** left, we stop. However, more often we will simply continue -- i.e., we look for the next largest power $10^{1/2^k}$ we can divide what's left by to stay over **1**, updating what's left to be this quotient, and then doing this over and over.

Here, we discover we can divide our initial "left over" value of **7.532** first by $10^{1/2}$, and then by $10^{1/4}$ and $10^{1/8}$, while always keeping the quotient greater than **1**. However, division by an additional $10^{1/16}$ or $10^{1/32}$, or anything bigger than $10^{1/512}$ drops the quotient less than **1** -- so we don't do any of these divisions. Instead, we divide by $10^{1/1024}$.

The more divisions we make, the better our approximation will be -- however, let's see how to put just what we have together.

Note that we can now say $10^3 \cdot 10^{1/2} \cdot 10^{1/4} \cdot 10^{1/8} \cdot 10^{1/1024}$ is a slight underestimate of **7532**. (*We know this product is less than 7532 as we never let our aforementioned quotients get below 1.*)

As such, the (common) log of this big product will be a slight underestimate of **log 7532**, and

$$\begin{aligned}
 \log 7532 &\doteq \log(10^3 \cdot 10^{1/2} \cdot 10^{1/4} \cdot 10^{1/8} \cdot 10^{1/1024}) \\
 &\doteq \log 10^3 + \log 10^{1/2} + \log 10^{1/4} + \log 10^{1/8} + \log 10^{1/1024} \\
 &\doteq 3 + 1/2 + 1/4 + 1/8 + 1/1024 \\
 &= 3.8759765625
 \end{aligned}$$

Had we done more divisions than what we did above, we would have an even better approximation of

$$3 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{1024} + \frac{1}{2048} + \frac{1}{4096} + \frac{1}{8192} + \frac{1}{16384} + \frac{1}{65536} + \frac{1}{524288}$$

which is just a bit larger than **3.8766**.

Comparing this to what your calculator's `log` button produces (i.e., **3.87691031**), we see we did pretty good!

There are methods that will converge on the actual value of the logarithm in question much more quickly -- but these will require we learn a bit more first. As such, let us not worry about these (..yet! 😊)